

Penetration Testing Framework

Scan Types Documentation

Version 1.0

Generated: March 05, 2026

Complete guide to network scanning methodologies

Table of Contents

Section	Page
1. Overview of Scan Types	3
2. Quick Scan	4
3. Service Scan	6
4. Stealth Scan	8
5. Full Scan	10
6. Comparison Matrix	12
7. Use Case Examples	13
8. Best Practices	15

1. Overview of Scan Types

The Penetration Testing Framework provides four distinct scan types, each optimized for different scenarios and requirements. Choosing the right scan type is crucial for balancing speed, stealth, and thoroughness in your security assessments.

Scan Type	Speed	Ports	Stealth	Best For
Quick	■ Fast	100	Medium	Reconnaissance
Service	■ Medium	1,000	Low	Vulnerability Assessment
Stealth	■ Slow	1,000	High	IDS Evasion
Full	■ Very Slow	65,535	Low	Comprehensive Audit

2. Quick Scan

Description

Quick Scan is designed for rapid network reconnaissance. It scans only the top 100 most common ports, making it ideal for initial assessments and large-scale network inventories.

Technical Details

```
nmap -T4 -F [target]
```

Command Breakdown:

- **-T4**: Fast timing template (aggressive)
- **-F**: Fast mode (scan top 100 ports only)

Performance Metrics:

- Time per host: 5-30 seconds
- Network load: Low
- Accuracy: Good for common services

When to Use

■ Use For	■ Avoid For
Initial reconnaissance	Complete security audits
Large networks (100+ hosts)	Finding uncommon services
Time-sensitive assessments	Compliance requirements
Daily/weekly monitoring	Detailed penetration testing
Confirming host availability	Searching for backdoors

Example Use Case

Scenario: Office Network Inventory
Situation: New IT administrator needs overview of network devices
Goal: Quick identification of all active hosts and main services
Results: Completed scan of 50 hosts in 2 minutes, identified web servers, file shares, and remote access services
Outcome: Created initial network map for further investigation

3. Service Scan (Recommended)

Description

Service Scan is the most comprehensive option for vulnerability assessments. It performs version detection, runs default security scripts, and provides detailed information about each discovered service. This is the recommended scan type for most security assessments.

Technical Details

```
nmap -sV -sC -T4 [target]
```

Command Breakdown:

- **-sV**: Version detection (identifies exact service versions)
- **-sC**: Script scanning (runs NSE vulnerability scripts)
- **-T4**: Fast timing template

Performance Metrics:

- Time per host: 1-3 minutes
- Network load: Medium
- Accuracy: Excellent - detailed service information

Detects:

- Service versions (Apache 2.4.49, OpenSSH 7.4, etc.)
- Operating system information
- SSL/TLS certificates
- Common misconfigurations
- Known vulnerabilities (CVEs)

When to Use

■ Use For	■ Avoid For
Vulnerability assessments	Very large networks (use Quick first)
Security audits	When stealth is critical
Penetration testing preparation	Quick status checks
Compliance checks (PCI, HIPAA)	Time-sensitive reconnaissance
Incident response analysis	Routine daily monitoring

Example Use Case

Scenario: Annual Security Assessment

Situation: Client requires comprehensive vulnerability report

Goal: Identify all vulnerabilities with CVE mappings

Results: Found Apache 2.4.49 on port 80, matched to CVE-2021-41773 (Critical 9.8)

Outcome: Generated professional PDF report with exploitation recommendations

4. Stealth Scan

Description

Stealth Scan uses SYN scanning and slower timing to minimize detection by intrusion detection systems (IDS) and intrusion prevention systems (IPS). Ideal for red team engagements and testing security monitoring effectiveness.

Technical Details

```
nmap -sS -T2 [target]
```

Command Breakdown:

- **-sS**: SYN scan (half-open scanning, stealthier)
- **-T2**: Polite timing (slower, harder to detect)

Performance Metrics:

- Time per host: 5-10 minutes
- Network load: Low (spread over time)
- Detection risk: Low

Evasion Techniques:

- Half-open TCP connections (never completes handshake)
- Slower packet rate (mimics normal traffic)
- Often bypasses simple IDS rules
- Reduces logging on target systems

SYN Scan vs Normal Connection

Normal TCP Connection:

Client → SYN → Server

Server → SYN-ACK → Client

Client → ACK → Server (connection established and logged)

SYN Scan:

Client → SYN → Server

Server → SYN-ACK → Client

Client → RST → Server (connection aborted, often not logged)

When to Use

■ Use For	■ Avoid For
Red team engagements	Routine security monitoring

IDS/IPS evasion testing	Your own systems (stealth unnecessary)
Production environment scanning	Time-sensitive assessments
Testing security monitoring	Quick network checks
Adversarial simulations	Large-scale scans (too slow)

Example Use Case

Scenario: Red Team Assessment

Situation: Hired to test company's security monitoring capabilities

Goal: Scan network without triggering IDS alerts

Results: Successfully scanned 30 hosts over 4 hours without detection

Outcome: Demonstrated IDS blind spots; client improved monitoring rules

5. Full Scan (Comprehensive)

Description

Full Scan is the most thorough option, scanning all 65,535 TCP ports with complete service detection, OS fingerprinting, and aggressive script scanning. Required for PCI-DSS compliance and recommended for critical infrastructure audits.

Technical Details

```
nmap -sV -sC -A -T4 -p- [target]
```

Command Breakdown:

- **-sV**: Version detection
- **-sC**: Script scanning
- **-A**: Aggressive mode (OS detection, traceroute)
- **-T4**: Fast timing
- **-p-**: Scan ALL 65,535 ports

Performance Metrics:

- Time per host: 10-20 minutes
- Network load: High
- Accuracy: Complete - finds everything

Discovers:

- Backdoors on unusual ports (31337, 4444, etc.)
- Custom applications (8080, 9000, etc.)
- Development servers accidentally exposed
- Legacy services on uncommon ports
- Everything Quick/Service scans miss

When to Use

■ Use For	■ Avoid For
Critical servers (DB, DC)	Large networks (too time-consuming)
Suspected compromises	Regular monitoring (overkill)
PCI-DSS compliance (required)	Quick assessments
Pre-decommissioning checks	Low-risk systems
APT investigations	Daily/weekly scans

Example Use Case

Scenario: Suspected Server Compromise

Situation: Database server exhibiting unusual network activity

Goal: Find any backdoors or unusual services

Results: Discovered backdoor listening on port 54321 (not found in Quick scan)

Outcome: Confirmed breach, initiated incident response, backdoor removed

6. Detailed Comparison Matrix

Feature	Quick	Service	Stealth	Full
Ports Scanned	100	1,000	1,000	65,535
Version Detection	No	Yes	No	Yes
OS Detection	No	Yes	No	Yes
NSE Scripts	No	Yes	No	Yes
Stealth Level	Medium	Low	High	Low
Speed	Fast	Medium	Slow	Very Slow
Network Load	Low	Medium	Low	High
Finds Backdoors	Unlikely	Maybe	Maybe	Yes
CVE Matching	Limited	Excellent	Limited	Excellent
Time/Host	30 sec	2 min	8 min	15 min

7. Real-World Use Case Examples

By Industry

Healthcare (HIPAA Compliance)

- Daily monitoring: Quick Scan (fast, minimal disruption)
- Monthly audit: Service Scan (detailed vulnerability assessment)
- Critical systems: Full Scan (comprehensive analysis)
- Production scans: Stealth Scan (avoid service disruption)

E-commerce (PCI-DSS Compliance)

- Quarterly requirement: Full Scan (mandated by PCI-DSS)
- Weekly monitoring: Service Scan (ongoing security)
- Peak hours: Stealth Scan (minimize customer impact)
- After updates: Service Scan (verify changes)

Corporate IT

- Daily checks: Quick Scan (identify new devices)
- Monthly audits: Service Scan (vulnerability management)
- Incident response: Full Scan (find backdoors)
- Testing IDS: Stealth Scan (verify monitoring)

Penetration Testing

- Initial recon: Quick Scan (fast mapping)
- Vulnerability assessment: Service Scan (find exploits)
- Evasion testing: Stealth Scan (bypass defenses)
- Final sweep: Full Scan (ensure nothing missed)

8. Best Practices

Scanning Strategy

Phase 1: Discovery (Quick Scan)

- Goal: Identify active hosts
- Time: 5 minutes for entire network
- Output: List of live systems

Phase 2: Identification (Service Scan)

- Goal: Detailed service information
- Time: 2 minutes per host
- Output: Services, versions, vulnerabilities

Phase 3: Deep Dive (Full Scan on Critical Systems)

- Goal: Complete audit of key servers
- Time: 15 minutes per critical host
- Output: Comprehensive security assessment

Phase 4: Stealth Test (Optional)

- Goal: Test security monitoring
- Time: 10 minutes
- Output: Verify IDS/IPS effectiveness

General Best Practices

■ Do	■ Don't
Start with Quick scan first	Full scan everything (too slow)
Use Service for vulnerabilities	Skip version detection
Full scan critical systems only	Scan during peak hours
Test with Stealth occasionally	Ignore stealth completely
Document scan choices	Forget to get authorization
Schedule appropriately	Scan production without notice

Decision Tree

Choosing the Right Scan Type:

1. Do you need results in under 1 minute?

→ YES: Use Quick Scan

→ NO: Continue to question 2

2. Is stealth/IDS evasion important?

→ YES: Use Stealth Scan

→ NO: Continue to question 3

3. Is this a comprehensive security audit?

→ YES: Is it a critical server?

→ YES: Use Full Scan

→ NO: Use Service Scan

→ NO: Use Service Scan

Compliance Requirements

PCI-DSS (Payment Card Industry):

- Requirement: Quarterly vulnerability scans
- Scan type: Full Scan (all ports mandatory)
- Frequency: Every 90 days minimum

HIPAA (Healthcare):

- Requirement: Regular risk assessments
- Scan type: Service Scan (standard) or Full Scan (critical systems)
- Frequency: Annually or after significant changes

SOC 2:

- Requirement: Security monitoring
- Scan type: Service Scan
- Frequency: Monthly or quarterly

End of Documentation

Penetration Testing Framework
Scan Types Documentation v1.0

For questions or support, consult the framework documentation
or refer to the online help system.

■■ **Legal Notice:** Always obtain proper authorization before
conducting security scans. Unauthorized scanning may be illegal.